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10/9/23 Mathematics (1)

1 3

2 No sol^n

3 10 units

4 Proportional

5 $\frac{-1}{2}$

6 x^3y^3

7 Sum of zeroes = $\frac{1}{2}$ Their product

$\alpha+\beta = \frac{1}{2}\alpha\beta$

$\frac{-b}{a} = \frac{1}{2}\left(\frac{c}{a}\right)$

$\frac{1}{2} = \frac{k-4}{8}$

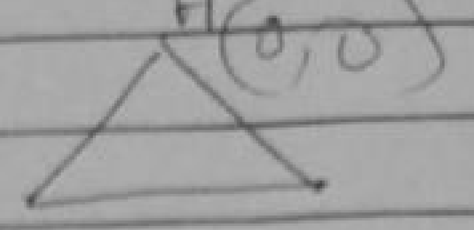
$8 = 2k-8$

$2k = 16$

$k = 8$

chandra's

8 $AB^2 = AC^2$ (because its an equilateral $\triangle^{le}$)



$(3-0)^2 + (\sqrt{3}-0)^2 = (3-0)^2 + (x-0)^2$

$\Rightarrow (\sqrt{3}-0)^2 = (x-0)^2$

$x = \sqrt{3}$ //

9 $2px + 3y = 7$

$2x + y = 6$

$\frac{2p\cancel{x}}{2\cancel{x}} \neq \frac{3}{1}$ (Given)

$2p \neq 6$

$p \neq 3$ //

16 PD // BC

$AP/PB = AD/DC$ ~~(Given)~~ (B.P.T) $\rightarrow$ ①

$AD/DC = CE/BE$ (Given) $\rightarrow$ ②

$\frac{AP}{PB} = \frac{CE}{BE}$ (From ① & ②)

by Converse of BPT

DE // AC

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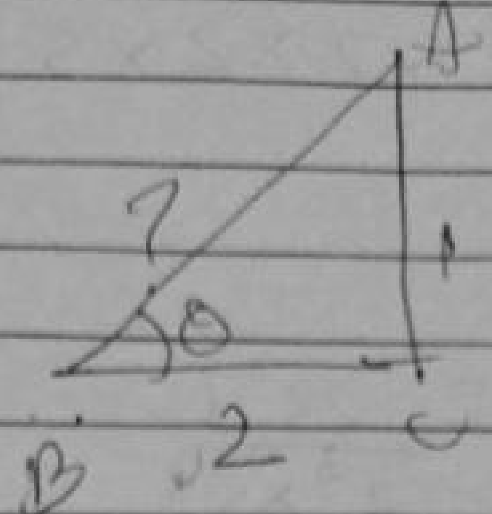
$\tan\theta = \frac{opp}{adj}$



$AB^2 = AC^2 + BC^2$
$= 1 + 4$
$= 5$
$AB = \sqrt{5}$

$\sin\theta = \frac{opp}{hypotenuse}$

$= \frac{1}{\sqrt{5}}$

~~$\cos\theta = \frac{adjacent}{hypotenuse}$~~ $\cos\theta = \frac{adj}{hypotenuse}$

$= \frac{2}{\sqrt{5}}$

$\frac{1}{2}\sin\theta\cos\theta = \frac{1}{5}$

$\frac{1}{\cancel{2}} \times \frac{1}{\sqrt{5}} \times \frac{\cancel{2}}{\sqrt{5}} = \frac{1}{5}$

$\frac{1}{5} = \frac{1}{5}$ Hence proved

12 $5000 = 2 \times 2 \times 2 \times 5 \times 5 \times 5 \times 5$

$\cancel{= 2^3 \times 5^4}$

$\frac{257}{5000} = \frac{257}{2^3 \times 5^4}$

$\Rightarrow \frac{257}{2^3 \times 5^4} \times \frac{2}{2}$

$\frac{257 \times 2}{5+4}$

$\frac{514}{(10)^4}$

$= 0.0514$

13 $\sqrt{3} = \frac{p}{q}$ Let us first assume that $\sqrt{3}$ is rational and is in the form p/q and $q \neq 0$ (p/q are coprime integers)

Square on both sides

$3 = \frac{p^2}{q^2}$

$3q^2 = p^2$

$\Rightarrow$ 3 is a factor of p^2

$\therefore$ 3 is a factor of p also

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$p = 3k$

Square on Both sides

$p^2 = 9k^2$

$3q^2 = 9k^2$

$q^2 = 3k^2$

$\Rightarrow$ 3 is a factor of q^2
$\therefore$ 3 is a factor of q also

3 is now the factor of both p and q

Thus contradicts our statement that p,q are coprime integers

$\therefore \sqrt{3}$ is an irrational no.

16 Given $\Delta ABE \cong \Delta ACD$

$AB = AC$ (C.P.C.T) $\rightarrow$ (1)
~~BE = CD (C.P.C.T) → 2~~

~~$\Rightarrow \frac{AB}{B}$~~ $AE = AD$ (C.P.C.T) $\rightarrow$ (2)

Dividing (2) by (1)

$\frac{AD}{AB} = \frac{AE}{AE}$

(6)

In ΔADE & ΔABC

$\angle A = \angle A$ (common)

$\frac{AD}{AB} = \frac{AE}{AC}$ (proved)

By SAS criterian

$\Delta ADE \sim \Delta ABC$ // Hence proved

17 $$\frac{\frac{\sin^2\theta}{\cos^2\theta}}{\frac{\sin^2\theta}{\cos^2\theta} - 1} + \frac{\frac{1}{\sin^2\theta}}{\frac{1}{\cos^2\theta} - \frac{1}{\sin^2\theta}}$$

$$\frac{\frac{\sin^2\theta}{\cos^2\theta}}{\frac{\sin^2\theta - \cos^2\theta}{\cos^2\theta}} + \frac{\frac{1}{\sin^2\theta}}{\frac{1}{\cos^2\theta} - \frac{1}{\sin^2\theta}}$$

$$\frac{\sin^2\theta}{\cos^2\theta} \times \frac{\cos^2\theta}{\sin^2\theta - \cos^2\theta} + \frac{\frac{1}{\sin^2\theta}}{\frac{\sin^2\theta - \cos^2\theta}{\sin^2\cos^2\theta}}$$

$$\frac{\sin^2\theta}{\sin^2\theta - \cos^2\theta} + \frac{\cancel{\sin^2\theta}\cos^2\theta}{\cancel{\sin^2\theta}(\sin^2\theta - \cos^2\theta)}$$

$$\frac{\sin^2\theta + \cos^2\theta}{\sin^2\theta - \cos^2\theta} = \frac{1}{\sin^2\theta - \cos^2\theta}$$

$= R.H.S$

~~Hence proved~~ Hence proved

19 Let the fraction be $\frac{x}{y}$

$\frac{2x}{y+1} = 1$ (Given)

$2x = y + 1$

$\frac{x+4}{2y} = \frac{1}{2} \Rightarrow x + 4 = y$

$2x - y - 1 = 0$

$x - y + 4 = 0$

$(-) \quad (+) \quad (-)$

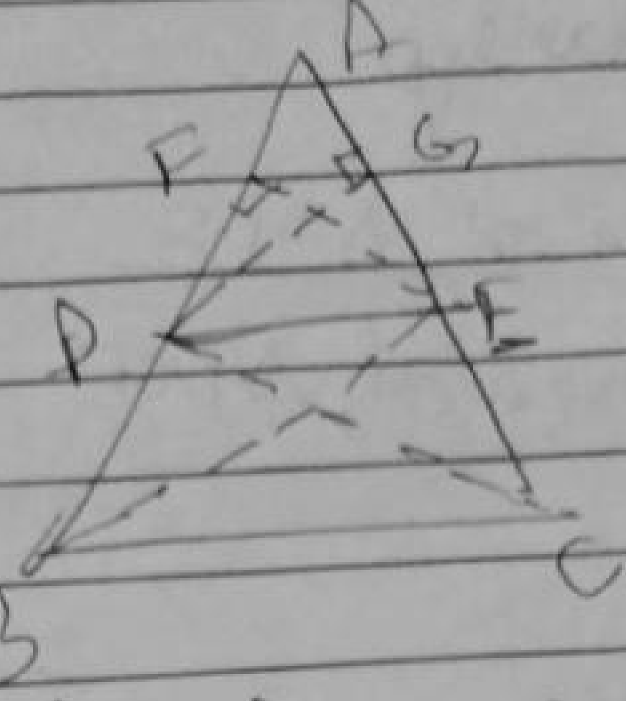
$x = -5$

~~$x + 5 + 4 = 0$~~ ~~$-5 - y + 4 \neq 0$~~

$-1-$

$y = -9$

$\therefore$ The fraction is $\frac{5}{9}$



(8)

20 BPT states that, if a line parallel to a side of a triangle it intersects the other 2 sides then the line divides it into equal proportion

ABC be a Δ^{le}

$BC \parallel DE$

To prove $\frac{AD}{DB} = \frac{AE}{EC}$

Join BE, CD

construction: $EF \perp AB$, $DG \perp CA$

$\therefore$ Area of $\Delta ADE = \frac{1}{2} \times$ base $\times$ height

$= \frac{1}{2} \times AD \times EF$

Area of $\Delta DBE = \frac{1}{2} \times AE \times DG$

$\frac{\text{Area of } \Delta ADE}{\text{" } \Delta DBE} = \frac{AD}{DB} \rightarrow (1)$

chandra's

Similarly

$$\frac{\text{area of } \Delta ADE}{\text{area of } \Delta DCE} = \frac{\frac{1}{2} \times AE \times DG}{\frac{1}{2} \times EC \times DG}$$

$$= \frac{AE}{EC}$$

ΔDBE and ΔDCE have the same base b/w same parallels

$\cancel{\therefore \frac{AD}{AB} = \frac{AE}{AC}}$ $\quad \therefore \frac{AD}{DB} = \frac{AC}{\cancel{AE}C}$

21 $$\frac{\frac{1}{2} + 2\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{\sqrt{3}}{1}\right)^2}{\frac{1}{2}\left(\frac{1}{\cancel{4}}\right) + \left(\frac{\sqrt{3}}{2}\right)^2 + 1^2}$$

$$\frac{\frac{1}{2} + \frac{2}{2} + \cancel{8}3}{\frac{1}{2} + \frac{3}{4} + 1}$$

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$\frac{1}{2} + \frac{2}{2} + 3$ (11)

$\frac{1}{2} + \frac{3}{4} + 1$

$\frac{3}{2} + 3$

$\frac{1}{2} + \frac{3}{4} + 1$

$\frac{\frac{3+6}{2}}{\frac{2+3+4}{4}} \Rightarrow \frac{\cancel{9}}{\cancel{2}} \times \frac{\cancel{4}2}{\cancel{9}}$

$\therefore \frac{\sin 30 + 2\cos^2 45 + \tan^2 60}{\frac{1}{2}\cot 45^\circ + \cos^2 30 + \tan^2 45} = 2$

23 Time covered by A (One round) $= \frac{1980}{330} = 6$ min

Time covered by B (One round) $= \frac{1980}{198} = 10$ min

" " C (One round) $= \frac{\cancel{1980}9}{\cancel{220}1} = 9$ min

Father's age be x
24 $(2y + x = 70) \times 2$ son's age be y
$2x + y = 95$

(11)

$4y + 2x = 140$
$y + 2x = 95$

$\Rightarrow 3y = 45$
$y = 15$
$15 + 2x = 95$
$2x = 80$

$x = 40$

Father's age 40
son's age 15

25 (6, 4) and (1, -7) be divided by the point on x axis (x, 0)

let us assume the ratio to be k : 1

$$\frac{-7k + 4}{k + 1} = 0 \quad \text{(Section formula)}$$

$-7k + 4 = 0$

$k = \frac{4}{7}$

$m : n \Rightarrow \frac{4}{7} : 1 \Rightarrow 4 : 7$

$m + n = 4 + 7 = 11$

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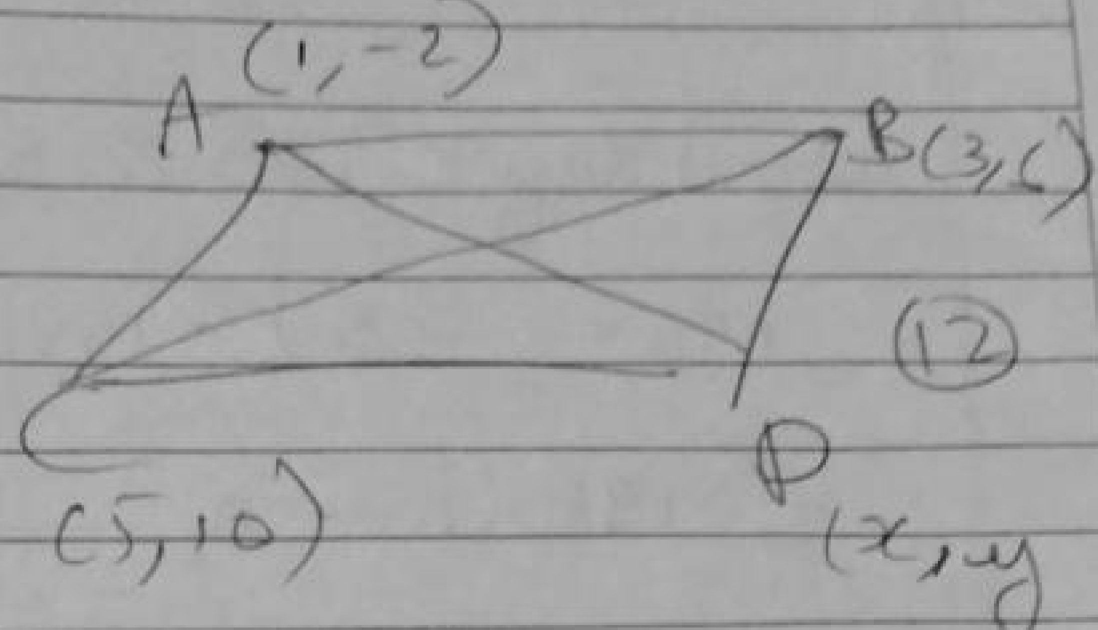
W.K.T
diagonals
of a $\parallel^{gm}$ bisect
each other



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So the mid point of AC is same as mid point of BD

$$\underset{AC}{\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)} = \underset{BD}{\left(\frac{3+x}{2}, \frac{6+y}{2}\right)}$$

$$\left(\frac{1+5}{2}, \frac{-2+10}{2}\right) = \left(\frac{3+x}{2}\right), \left(\frac{6+y}{2}\right)$$

$$\frac{6}{2}, \frac{8}{2} = \frac{3+x}{2}, \frac{6+y}{2}$$

$\cancel{3+x}$ $3+x=6$; $6+y=8$

$x=3$; $y=2$

So ~~DC~~ Fourth vertex's coordinates are (3,2)

23rd question continuation

23 A,B,C meet after time t and completed n_1, n_2, n_3 rounds respectively

$330t = x + 1980n_1 \rightarrow (1)$

$198t = x + 1980n_2 \rightarrow (2)$

$\ldots = x + 1980n_3 \rightarrow (3)$

(13)

substituting x into (2) & (3)

$198t = 330t - 1980n_1 + 1980n_2$

$1980(n_1 - n_2) = 132t$

$t = 15(n - n_2)$

Also, $220t = 330t - 1980n_1 + 1980n_3$

$1980(n_1 - n_3) = 110t$

$t = 18(n_1 - n_3)$

Since t is same and $n_1 - n_3$ and $n_1 - n_2$ can be anywhole no.

LCM of 15 and 18 is 90

A, B, C will meet after 90 mins

28 i)

D of PR = $\frac{8-2}{2}$, $\frac{3-5}{2}$

$= \frac{6}{2}, \frac{-2}{2}$

$= 3, -1$

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Case Relative speed $x-y$ (same direction

29 Dist. $(100) = (x-y)5$ (14)

$x-y=20$

$x+y$ (opposite direction)

$x+y=100$

$x-y=20$
$x+y=100$

$2x=120$

$x=60$ km/h

$y=40$ km/h

30 $\Delta ABD \sim \Delta PQD$ (corresponding sides are parallel)

$\frac{x}{z} = \frac{BD}{QD} \Rightarrow \frac{1}{x} = \frac{QD}{z \times BD} \rightarrow (1)$

$\Delta CDB \sim PQB$ (corresponding sides are parallel)

$\frac{y}{z} = \frac{BD}{BQ} \Rightarrow \frac{1}{y} = \frac{BQ}{z \times BD} \rightarrow (2)$

$\frac{1}{x} + \frac{1}{y} = \frac{BQ+QD}{z \times BD} = \frac{1}{z}$ // Hence proved

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$\alpha + \beta = \frac{-3}{2}$

(15)

$\alpha \beta = \frac{7}{2}$

$\frac{1}{\alpha^2} + \frac{1}{\beta^2} \Rightarrow \frac{\alpha^2 + \beta^2}{\alpha^2 \beta^2}$

$$= \frac{\left(\frac{-3}{2}\right)^2 - \left(2 \times \frac{7}{2}\right)^2}{\left(\frac{7}{2}\right)^2}$$

$$= \frac{\frac{9}{4} - 7}{\frac{49}{4}}$$

$$= \frac{-18}{49}$$

Also $\frac{1}{(\alpha\beta)^2} = \frac{4}{49}$

$x^2 + \left(\frac{18}{49}\right)x + \frac{4}{49}$

$49x^2 + 18x + 4 = 0$ //

chandra's

16

15 $\frac{3}{5} = \frac{-a-1}{1-2a}$

$-5(a+1) = 3(1-2a)$

$a = 8$

$\frac{3}{5} = \frac{2b-1}{3b}$

$9b = 10b-5$

$b = 5$